

FuMOGAE: A Fuzzy Multi-Omics Graph AutoEncoder for Breast Cancer Subtype Classification

1 Supplementary Table

Table 1: Classification results of breast cancer subtypes using different omics data combinations

Omics Data	ACC	AUC	F1_weight
Clinical	0.704	0.9390	0.7197
CNA	0.400	0.6590	0.4034
GSE	0.616	0.7730	0.5961
DNA	0.440	0.7093	0.4331
miRNA	0.688	0.9217	0.6932
Mutation	0.496	0.7137	0.4928
Co-expression	0.472	0.7298	0.4699
Clinical + CNA	0.744	0.9422	0.7506
CNA + GSE	0.608	0.7692	0.5943
GSE + DNA	0.536	0.7893	0.5402
CNA + GSE + DNA	0.520	0.7923	0.5233
CNA + GSE + DNA + miRNA	0.696	0.9120	0.7029
CNA + GSE + DNA + miRNA + Mutation	0.792	0.9238	0.7913
CNA + GSE + DNA + miRNA + Mutation + Coexp	0.776	0.9178	0.7736

Table 2: Ablation study of FuMOGAE

Model Variant	ACC	AUC	F1_weighted
FuMOGAE (GAE + Contrastive + Choquet + Sugeno)	0.9040	0.9882	0.9043
GAE + Choquet	0.8880	0.9894	0.8855
GAE + Sugeno	0.8480	0.9846	0.8421

Table 3: Comparison of the proposed Hybrid (Choquet + Sugeno) model with existing methods for multi-class breast cancer subtype classification

Method	Accuracy	F1_Weighted	F1_Macro
SVM	0.617	0.627	0.535
RF	0.621	0.635	0.543
NN	0.649	0.633	0.584
NN_NN	0.699	0.679	0.611
NN_VCDN	0.687	0.671	0.609
GCN	0.730	0.721	0.525
GAT	0.733	0.725	0.552
MOGONET	0.712	0.717	0.614
MVGNN	0.735	0.725	0.636
FuMOGAE (Proposed)	0.9040	0.9043	0.8943

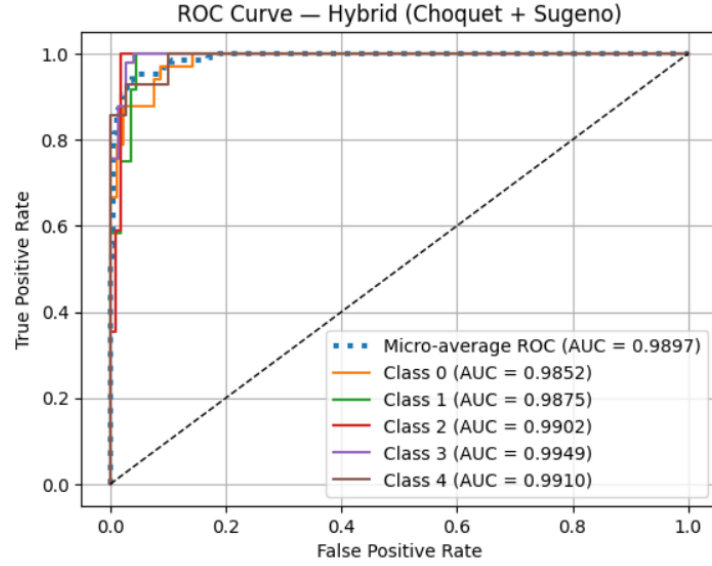


Fig. 1: Performance comparison of FuMoGAE with different fusion strategies.

Algorithm 1 FuMOGAE: Phase I – Multi-Modal Data Processing and Graph Construction

Require: Clinical, CNA, GSE, DNA, miRNA, Mutation, and Coexpression datasets

Ensure: Processed modality features and graph structures

- 1: **Step 1: Multi-Modal Data Loading**
- 2: Read modality-specific CSV datasets
- 3: Remove empty rows and invalid samples
- 4: Extract patient features and subtype labels
- 5: **Step 2: Label Encoding**
- 6: LumB=0, Normal=1, Her2=2, LumA=3, Basal=4
- 7: **Step 3: Mutation Data Processing**
- 8: Remove patient identifiers
- 9: Convert mutation profiles into binary values
- 10: Remove low-frequency mutation features:
- 11: $\sum x_j > 10$
- 12: Append subtype labels
- 13: **Step 4: Unified Data Preprocessing**
- 14: **for** each modality M **do**
- 15: Remove patient ID column
- 16: Convert features into numeric format
- 17: Replace missing values using feature-wise mean
- 18: Replace remaining NaN values with zero
- 19: **if** $M = \text{Clinical}$ **then**
- 20: Apply MinMax normalization
- 21: **else**
- 22: Apply Z-score standardization
- 23: **end if**
- 24: **end for**
- 25: **Step 5: Feature Selection**
- 26: **for** each modality M **do**
- 27: **if** $M = \text{Clinical}$ **then**
- 28: Skip feature selection
- 29: **else if** $M \in \{\text{Mutation, COEXP}\}$ **then**
- 30: Skip feature selection
- 31: **else**
- 32: **if** $M = \text{DNA}$ **then**
- 33: Apply Mutual Information prefiltering
- 34: Retain top 3000 informative features
- 35: **end if**
- 36: Apply Boruta feature selection
- 37: Apply mRMR optimization:
- 38: $\max(\text{Relevance} - \text{Redundancy})$
- 39: **end if**
- 40: **end for**
- 41: **Step 6: Feature Standardization**
- 42: **for** each modality M **do**
- 43: Apply StandardScaler normalization
- 44: **end for**
- 45: **Step 7: KNN Gaussian Graph Construction**
- 46: **for** each modality M **do**
- 47: Compute Euclidean distance matrix:
- 48: $D(i, j) = \|x_i - x_j\|_2$
- 49: Compute Gaussian similarity matrix:
- 50: $W(i, j) = \exp\left(\frac{-D(i, j)^2}{2\sigma^2}\right)$
- 51: Construct KNN graph with $k = 10$
- 52: Symmetrize adjacency matrix:
- 53: $W = \max(W, W^T)$
- 54: Remove self-loops
- 55: **end for**
- 56: **Step 8: Graph Visualization**
- 57: Visualize modality-specific graphs using NetworkX

Algorithm 2 FuMOGAE: Phase II – Multi-View Graph Autoencoder Learning

Require: Processed modality features and graph adjacency matrices

Ensure: Latent graph embeddings

1: **Step 9: Stratified 10-Fold Cross Validation**

2: Apply StratifiedKFold: $K = 10$

3: **for** each fold $f = 1 \rightarrow 10$ **do**

4: Split train and test samples

5: **Step 10: Fold-wise Graph Splitting**

6: Construct:

7: W_{train}, W_{test}

8: **Step 11: Graph Sparsification**

9: **for** each modality graph **do**

10: Retain strongest 10% edges

11: Ensure graph symmetry

12: Remove self-loops

13: **end for**

14: **Step 12: Graph Normalization**

15: Add identity matrix:

16: $\hat{A} = A + I$

17: Compute normalized graph:

18: $S = D^{-1/2} \hat{A} D^{-1/2}$

19: **Step 13: Train/Test Modality Split**

20: Split modality features into train and test subsets

21: **Step 14: Tensor Conversion**

22: Convert modality matrices into PyTorch tensors

23: **Step 15: Multi-View Graph Autoencoder Learning**

24: **for** each modality graph $\mathcal{G}_M$ **do**

25: Encode node features:

26: $Z = \text{ReLU}(SXW)$

27: Reconstruct graph:

28: $\hat{A} = \sigma(ZZ^T)$

29: Decode reconstructed features:

30: $\hat{X} = f_{\text{dec}}(Z, S)$

31: **end for**

32: **Step 16: Multi-Objective Loss Optimization**

33: Reconstruction loss:

34: $\mathcal{L}_{rec} = \sum ||X - \hat{X}||^2$

35: Contrastive loss:

36: $\mathcal{L}_{con}$

37: Classification loss:

38: $\mathcal{L}_{cls}$

39: Graph reconstruction loss:

40: $\mathcal{L}_{graph} = \sum ||A - \hat{A}||^2$

41: Total optimization loss:

42: $\mathcal{L} = \mathcal{L}_{rec} + 0.1\mathcal{L}_{con} + \mathcal{L}_{cls} + 0.5\mathcal{L}_{graph}$

43: Train using Adam optimizer

44: Apply gradient clipping and early stopping

45: **Step 17: Latent Embedding Extraction**

46: Extract modality-specific embeddings:

47: $Z_1, Z_2, \dots, Z_n$

48: **Step 18: Embedding Fusion**

49: Concatenate modality embeddings:

50: $Z = [Z_1 || Z_2 || \dots || Z_n]$

51: **Step 19: Embedding Normalization**

52: Apply StandardScaler normalization

53: Apply L2 normalization:

54: $Z = \frac{Z}{||Z||_2}$

55: **end for**

Algorithm 3 FuMOGAE: Phase III – Hybrid Choquet–Sugeno Fusion-Based Classification

Require: Normalized latent embeddings

Ensure: Final subtype prediction and evaluation metrics

1: **Step 20: Classifier Training**

2: Apply SMOTE for class balancing

3: Train:

4: {RF, FFNN, MLP}

5: Obtain predicted class probabilities

6: **Step 21: Fuzzy Measure Construction**

7: Compute classifier reliability scores

8: Construct fuzzy measure μ

9: **Step 22: Choquet Integral Fusion**

10: Aggregate classifier probabilities:

11: $C_\mu(f) = \sum (f_{(i)} - f_{(i-1)})\mu(A_{(i)})$

12: **Step 23: Sugeno Integral Fusion**

13: Compute Sugeno aggregation:

14: $S_\mu(f) = \max[\min(f_i, \mu_i)]$

15: **Step 24: Hybrid Choquet–Sugeno Fusion**

16: Compute adaptive fusion weight:

17: $\alpha = \frac{\text{Conf}_{Choquet}}{\text{Conf}_{Choquet} + \text{Conf}_{Sugeno}}$

18: Compute final fused probability:

19: $P = \alpha P_{Choquet} + (1 - \alpha) P_{Sugeno}$

20: Predict subtype:

21: $\hat{y} = \arg \max(P)$

22: **Step 25: Performance Evaluation**

23: Compute:

- Accuracy
- Macro F1-score
- Weighted F1-score
- Multi-class AUC

24: **Output**

25: Save:

- Fold-wise metrics
 - Subtype-wise performance tables
 - Confusion matrices
 - Mean $\pm$ standard deviation across 10 folds
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